

Yuxuan Huang

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EDUCATION

Columbia University

Master of Science in Mechanical Engineering (Robotics Concentration)

Rensselaer Polytechnic Institute

Bachelor of Science in Mechanical Engineering

New York, NY

Expected Dec 2026

Troy, NY

May 2025

PUBLICATION

Jin, A., Huang, Y., Pogorelyuk, L.

Deep-learning-based displacement measuring interferometry for non-centered and irregular fringes.

Proceedings of SPIE, 2025. DOI: <https://doi.org/10.1117/12.3062051>

TECHNICAL SKILLS

Programming & Data

- Python, MATLAB; Data analysis, visualization, machine learning; Algorithm development, debugging

Robotics & Controls

- Robot kinematics and dynamics; Gait generation and trajectory planning; PID control, motion control

CAD & Mechanical Design

- SolidWorks, Siemens NX, CATIA, Fusion 360; Mechanical design, tolerance analysis, GD&T

Manufacturing & Prototyping

- CNC machining, laser cutting; Additive manufacturing, sheet metal fabrication; Rapid prototyping and test validation
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PROJECT EXPERIENCE

Quadruped Robot System Design

New York, NY

Robotics Engineer

Sep 2025 – Dec 2025

- Designed, fabricated, and validated a servo-driven quadruped robot
 - Implemented multiple gait patterns with manual control algorithm and reinforcement learning, achieving stable locomotion
 - Integrated real-time servo control and feedback protection, improving motion stability and system reliability
 - Performed iterative mechanical and control tuning based on experimental testing results
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Electrochemical Machining (ECM) System Development

New York, NY

Manufacturing / Motion System Designer

Sep 2025 – Dec 2025

- Designed and built a benchtop electrochemical machining system for precision metal material removal
 - Developed and tested a functional prototype, meeting performance and operational criteria
 - Evaluated motion control, tool positioning, and process stability through experimental trials
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Motorized Caster Wheel Module

Troy, NY

Mechanical Product Designer

Feb 2023 – May 2023

- Developed a motorized auxiliary mobility module for logistics carts to reduce manual handling workload
 - Conducted design benchmarking and feasibility analysis of existing motorized steering systems
 - Fabricated a working prototype using sheet-metal parts and 3D-printed components
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WORK EXPERIENCE

BHTC (Shanghai) Co., Ltd. (Behr-Hella Thermo-Control)

Shanghai, China

Mechanical Design Engineer Trainee

Jun 2025 – Aug 2025

- Participated in automotive interior display concept development for production-oriented systems
 - Translated customer requirements into engineering specifications to support concept evaluation
 - Produced preliminary CAD models to assist downstream product development
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Rensselaer Polytechnic Institute

Troy, NY

Undergraduate Researcher, CubeSat Simulation Group

Jan 2024 – May 2025

- Developed a 3D position-tracking system for CubeSat simulation and visualization
 - Designed and executed data acquisition and processing pipelines for simulation analysis
 - Optimized machine learning models with research team to improve estimation accuracy
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CERTIFICATIONS

- Six Sigma Green Belt, Institute of Industrial and Systems Engineers (IISE)